

GroAurum

Phase 2 Sprint 4 - Production CRUD Foundation

Real Supabase backend integration

Generated: July 16, 2026

No UI redesign - repository architecture preserved

1. Sprint Goal

Connect GroAurum admin data access to the real Supabase backend. Extend the existing repository architecture with production CRUD (create, update, softDelete), Zod validation, React Query mutations, additive schema migrations, RLS for READ_ONLY, and a Sprint 4 seed. No new screens and no UI redesign.

2. Architecture (kept)

- packages/data - ReadRepository + CrudRepository contracts, query keys, QueryState
- packages/api - adapters, MemoryCache, DomainCrudRepositories, CrudServices
- packages/validation - shared Zod schemas
- packages/api-client - typed Supabase client + extended Database types
- Pages continue to load view-model data via repositories (mock sources)
- Mutations go through CrudServices -> domain repositories -> Supabase

3. Database ERD (logical)

```

profiles (user_profiles)
|
service_areas ---- operational_locations (warehouse)
|
shops (customers) ---- customer_addresses
|
orders ---- order_lines (order_items)
|
delivery_routes ---- route_stops

categories ---- products ---- product_images
|
|
skus ---- sku_prices (price_history)
|
inventory_balances (inventory)
|
inventory_movements

settings                      reports_snapshot (placeholder)
  
```

4. Migration List (0001-0023)

No.	Focus
0001-0016	Phase 2/2.5 core schema + RLS + grants
0017	Soft-delete deleted_at columns
0018	product_images
0019	customer_addresses
0020	settings + reports_snapshot
0021	Compat views (aliases)
0022	READ_ONLY staff_role
0023	RLS extensions for new tables + RO

Full filename map: supabase/migrations/MIGRATION_INDEX.md

5. Table Descriptions (Sprint 4 additions)

- product_images - normalized media rows; products.image_urls retained
- customer_addresses - extra shop delivery addresses; shops.delivery_* primary
- settings - key/jsonb admin configuration store
- reports_snapshot - placeholder for future analytics payloads
- Compat views: user_profiles, customers, price_history, order_items, inventory
- Soft delete: deleted_at on catalogue, shops, routes, profiles, areas, locations

6. Repository Mapping

Repository	Physical table(s)
categories	categories
products	products (+ product_images)
skus	skus
prices	sku_prices / price_history
inventory	inventory_balances + movements
customers	shops / customers view
orders	orders + order_lines
salesmen	profiles (SALESMAN role)
delivery	delivery_routes
settings	settings

7. CRUD Mapping

Each domain repository implements list, getById, search, subscribe, create, update, softDelete. Pricing is append-only (create + close). Inventory updates write ADMIN_ADJUSTMENT movements. Orders softDelete cancels (status=CANCELLED).

- CrudServices validates with Zod then calls repositories
- Products / Categories / Inventory / Pricing / Customers / Orders / Salesmen / Delivery / Settings services generated
- React Query mutations in apps/admin-web/src/data/mutations.ts
- Invalidation via queryKeys + AdminDataService.invalidate + MemoryCache
- Optimistic soft-delete for products (rollback on error)

8. RLS Overview

- Existing policies: ADMIN / SALESMAN / DELIVERY / CUSTOMER (Phase 2.5)
- NEW: READ_ONLY role - select on catalogue + settings/reports; no writes
- product_images / customer_addresses / settings / reports_snapshot policies
- Admin write remains is_admin(); salesman may insert customer_addresses for assigned shops
- App RBAC (packages/auth) stays finer-grained; DB maps to staff_role

9. Seed Overview

```
supabase/seed/sprint4_seed.sql
pnpm db:seed:sprint4
```

Includes: 2 categories, 2 products, 2 SKUs, prices, inventory, 1 warehouse, 1 admin, 2 salesmen, 5 customers, 2 sample orders, settings + reports_snapshot placeholder

10. Validation Overview

- Zod schemas in @groaurum/validation (category, product, sku, price, inventory, customer, order, salesman, delivery, settings)
- Shared helpers: uuid, pin code, money, quantity
- Services call schema.parse before repository writes

11. Future Realtime Points

- ReadRepository.subscribe / RealtimeBus already defined
- Domain repos publish INSERT/UPDATE/DELETE to bus after mutations
- createNoopRealtimeBus today - swap for supabase.channel postgres_changes
- Target tables: orders, inventory_balances, delivery_routes, sku_prices
- UI should invalidate queryKeys on realtime events (hooks prepared)

12. Adapter Switch

VITE_DATA_ADAPTER=mock | supabase
 VITE_SUPABASE_URL / VITE_SUPABASE_ANON_KEY

mock: view-model repos from fixtures; domain=null; CRUD unavailable
 supabase: domain CRUD live; UI still uses fixture view sources
 (no redesign) until view-model mappers land

13. Validation

Check	Result
@groaurum/data typecheck	Pass
@groaurum/validation typecheck + test	Pass
@groaurum/api typecheck + test	Pass
@groaurum/admin-web typecheck	Pass
UI redesign / new screens	None
Live realtime channels	Interfaces + publish hooks only

14. Intent

Sprint 4 establishes production database CRUD on the existing repository architecture so admin mutations and future view-model mappers can plug into stable Supabase-backed contracts without rewriting modules.